ENTERPRISE LAN

Designing a Secure and Fault Tolerant Enterprise Network

Capstone Final Presentation

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THE PROBLEM

Flat networks are easy to set up and impossible to defend

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compromised host

on a flat LAN exposes every other host on the same broadcast domain

Why it matters for SMBs

- **No internal trust boundaries**
  users, servers, and admin all share one segment
- **No defense in depth**
  perimeter is the only control
- **Hard to audit**
  no choke points to log inter zone traffic
- **Poor blast radius**
  lateral movement is trivial

THE APPROACH

Segmentation, default deny, and explicit policy — on commodity tools

Layered architecture

Four security zones, each its own VLAN and trust boundary

Firewall enforced policy

Inter zone traffic must traverse pfSense; nothing bypasses it

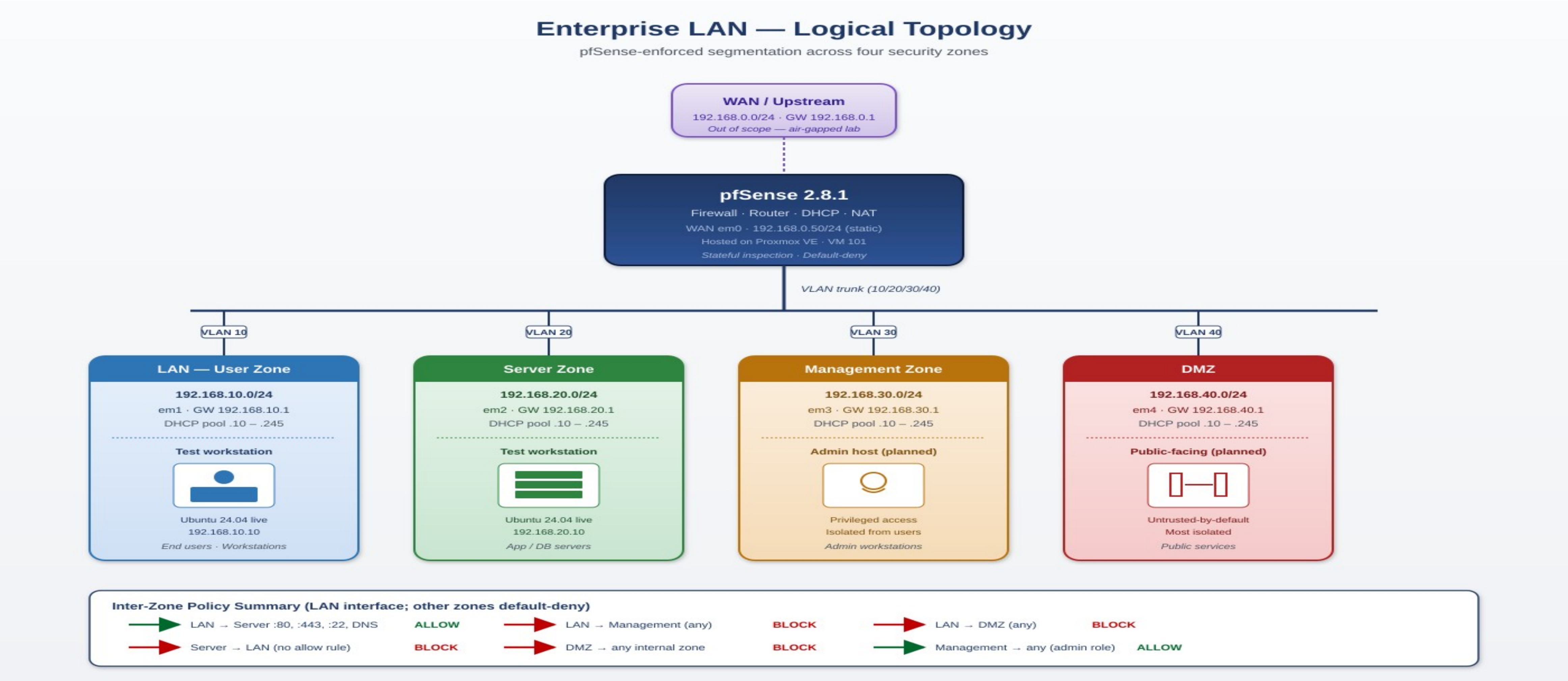
Default deny per zone

Zones are silent until policy explicitly opens a path (zero trust)

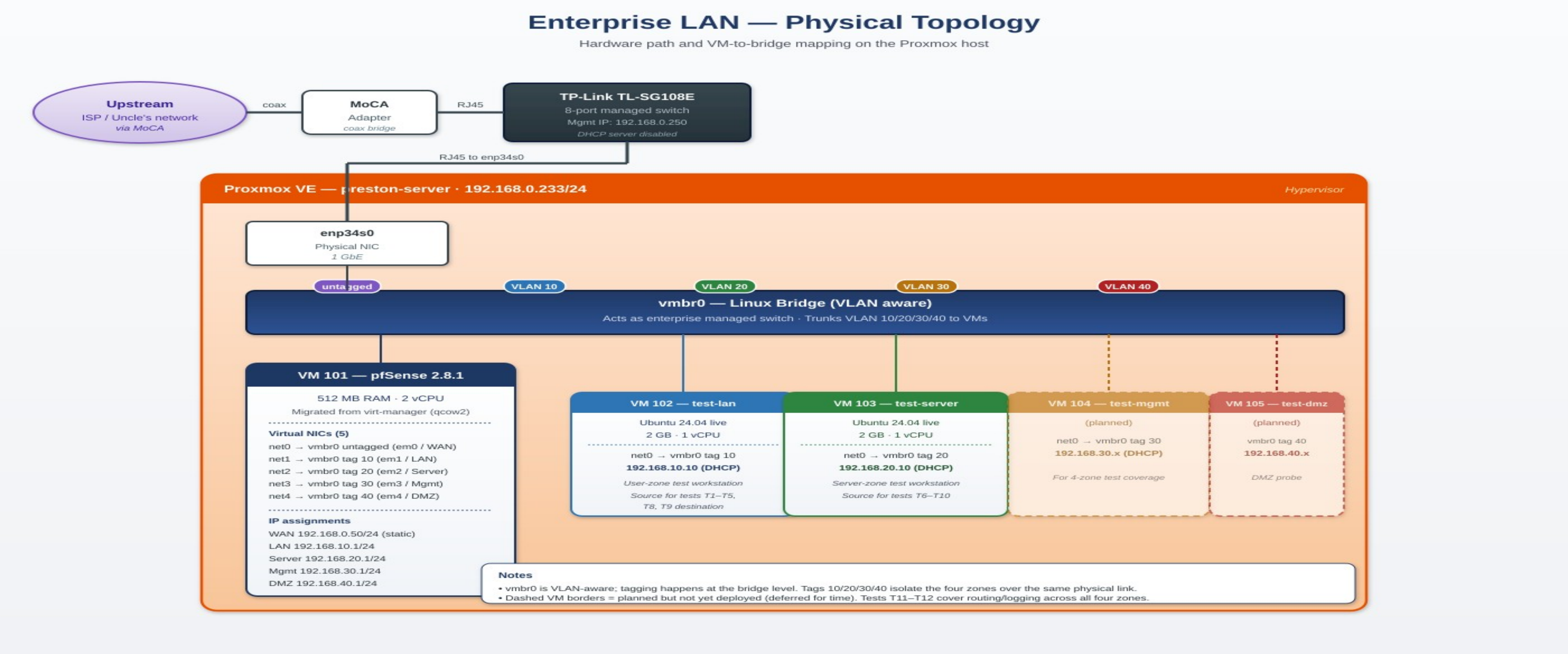
Open source stack

pfSense + Proxmox VE on commodity hardware — minimal cost, low complexity

Logical topology — four zones behind one firewall



Physical topology — one Proxmox host, one VLAN aware bridge



ARCHITECTURE

Four zones, four subnets, four DHCP scopes, four policies

Zone	Subnet	Gateway	DHCP Pool	Purpose
LAN — User	192.168.10.0/24	192.168.10.1	.10–.245	End user workstations
Server	192.168.20.0/24	192.168.20.1	.10–.245	App / DB servers
Management	192.168.30.0/24	192.168.30.1	.10–.245	Admin workstations (privileged)
DMZ	192.168.40.0/24	192.168.40.1	.10–.245	Public facing services (untrusted)
WAN (lab)	192.168.0.0/24	192.168.0.50 (static)	n/a	Upstream — air-gapped scope

POLICY

LAN ruleset — explicit allows, explicit blocks, default deny

#	Action	Source	Destination	Port	Description
1	Pass	LAN subnets	Server net	TCP 80, 443	HTTPS to Server
2	Pass	LAN subnets	Server net	TCP 22	SSH to Server
3	Pass	LAN subnets	Server VM (.10)	TCP/UDP 53	DNS
4	Block	LAN subnets	Mgmt net	*	Block users → Mgmt
5	Block	LAN subnets	DMZ net	*	Block users → DMZ
6	Pass	LAN subnets	any	*	Catch all (future internet)
—	Block	any	any	*	Default deny (implicit)

Server, Management, and DMZ interfaces have no allow rules → implicit default-deny.

Rules evaluate top down. First match wins.

What was actually built

pfSense 2.8.1

VM 101 · 5 NICs · static WAN · Kea DHCP · stateful policy

VLAN aware bridge

vmbr0 trunks tags 10/20/30/40 to pfSense and test workstations

LAN test workstation

Ubuntu 24.04 live · 192.168.10.10 · DHCP-assigned

Server test workstation

Ubuntu 24.04 live · 192.168.20.10 · DHCP-assigned

Migration

Migrated from libvirt → Proxmox; preserved disk, MACs, interface map

Test artifacts

12-test matrix · firewall log captures · ip a / TCP screenshots

Methodology — measure behavior, not configuration

Approach

1. **DHCP delivery**

   ip a after reacquiring lease

2. **TCP reachability**

   bash /dev/tcp pseudo device — no extra tools needed on live ISO

3. **ICMP reachability**

   ping -c with packet loss inspection

4. **Enforcement evidence**

   pfSense Status → System Logs → Firewall, filtered to test source IP

Example — TCP test from LAN VM

```
$ timeout 3 bash -c \
    'exec 3<>/dev/tcp/192.168.20.1/443' \
    && echo OK || echo BLOCKED
OK

$ timeout 3 bash -c \
    'exec 3<>/dev/tcp/192.168.30.1/443' \
    && echo OK || echo BLOCKED
BLOCKED

$ ping -c 2 192.168.10.10
...
2 packets transmitted, 0 received,
100% packet loss
```

Results — 12 tests, all matching design intent

12	11	1	0
tests executed	PASS — match expected	NOTE — engineering finding	fail

Sample results

#	Source	Destination	Expected	Actual	Result
T1	LAN VM	Server :443	Allow	OK	**PASS**
T3	LAN VM	Mgmt :443	Block	Blocked	**PASS**
T6	Server VM	DHCP	Lease	.10/24	**PASS**
T8	Server VM	LAN host (ICMP)	Block	100% loss	**PASS**
T10	Server VM	LAN gw :443	Block	OK	**NOTE**

Three engineering insights from testing

Rule order matters

A permissive 'allow internet access' rule above the explicit blocks made segmentation silently fail. pfSense evaluates top-down → most-specific-first wins.

Fix: Reordered rules; segmentation now enforces.

Firewall as target tests mislead

Cross zone tests aimed at pfSense's own gateway IPs all returned OK because of anti-lockout behavior — even when policy should have blocked them.

Fix: Retargeted tests at actual VM hosts inside destination zones for accurate measurement.

Default-deny works as designed

Server, Mgmt, and DMZ interfaces have no allow rules. Server → LAN tests confirm hosts in those zones cannot initiate cross zone traffic.

Fix: Implements zero-trust without writing block rules.

Scope decisions and what comes next

Scoped Out — With Rational

Live WAN connectivity

Upstream lab gateway constraints — air gapped scope adopted

HA pfSense pair (CARP)

Single host hardware — service-level redundancy used instead

Centralized DNS / web

Deferred to focus on segmentation and policy enforcement core

Cisco 3750 switch integration

On hand; switching simulated via VLAN aware Proxmox bridge

Future Work

Integrate Cisco 3750

Real L2 access layer with Proxmox host as trunk uplink

Deploy HA pfSense pair

CARP with state synchronization for true firewall failover

Add Unbound + monitoring

Centralized DNS and uptime monitoring on a Server zone VM

Expand testing to all 4 zones

Mgmt and DMZ test workstations + load testing

THANK YOU

Questions?

Enterprise LAN Capstone

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